CANDRON ENERGY INC.

FREE RESOURCE · PROJECT TOOLKIT

Cold-Climate & Arctic Readiness Checklist

What to specify, test, ship, install, energize and maintain so switchgear, transformers and e-houses work at northern and Arctic sites.

A Specify for the Cold

B Factory Test & Shipping

C Receiving, Storage & Installation

D Energization in the Cold

E Winter Operation & Maintenance

General guidance only. Follow the manufacturer's instructions, the project specification and the authority having jurisdiction. References: CSA C22.1, CSA C22.2 equipment standards, CSA C57 / IEEE C57 transformer standards and NETA ATS.

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SHEET A · DESIGN AND PROCUREMENT

Specify for the Cold

Most cold-weather failures are decided on paper. State the site conditions plainly, and make every supplier confirm each item in writing before the order is placed.

PROJECT _____ EQUIPMENT / TAG _____ DATE _____ WITNESS _____

Site conditions on every specification

- [ ] Minimum and maximum design ambient stated
  For example -45 °C to +35 °C. Use the site record, not a regional average.
- [ ] Snow load, drifting, ice and wind stated
- [ ] Altitude, salt, dust and wildlife exposure stated where relevant
- [ ] Foundation conditions known: frost heave, permafrost or muskeg
- [ ] Access stated: all-season road, winter road window, barge or air

Equipment ratings

- [ ] Every device rated for the minimum ambient, or heated
  Relays, meters, PLCs, HMIs and chargers are often rated to -20 °C or -40 °C only.
- [ ] Breaker and switch mechanisms use lubricants rated for the low temperature
- [ ] Enclosure steel, hardware and hinges suitable for low-temperature impact
- [ ] Gaskets and seals rated for the low temperature, typically silicone
- [ ] Nameplates and labels mechanically fixed, not adhesive only

Heat and moisture control

- [ ] Space heaters in every compartment, with thermostat and humidistat
- [ ] Heaters fed from a reliable source, with a heater-failure alarm
- [ ] Battery compartment heated, or battery sized for cold capacity loss
- [ ] Hooded, snow-shielded vents and filters that can be changed in winter

Transformers

- [ ] Liquid-filled: fluid pour point checked against the minimum ambient
  Natural ester fluids have higher pour points than mineral oil. Ask for the cold-start guidance.
- [ ] Liquid-filled: tank and gauges suit the full temperature range, with level marks for cold
- [ ] Dry-type: cast coil considered where the unit may sit de-energized in the cold
- [ ] Cooling fans and controls rated for the cold, or locked out below a set temperature

Layout and remote-site support

- [ ] Equipment raised above expected snow depth; doors swing clear of drifts
- [ ] E-houses and walk-in gear have a vestibule or arctic entry
- [ ] Shipping splits sized for the road, barge or aircraft
- [ ] Spare parts list agreed, including heaters, thermostats, fuses and relays

SHEET B · AT THE MANUFACTURER, BEFORE RELEASE FOR SHIPMENT

Factory Test & Shipping

Add these to the normal factory acceptance test. Problems with heaters, seals and packaging are cheap to fix in the factory and very expensive at a fly-in site.

PROJECT _____ EQUIPMENT / TAG _____ DATE _____ WITNESS _____

At the factory test

- [ ] Every space heater and thermostat energized and checked
- [ ] Heater-failure and low-temperature alarms proven
- [ ] Low-temperature ratings of all devices confirmed against the bill of materials
- [ ] Cold-chamber or low-temperature operation tests witnessed, where specified
- [ ] Gaskets, vent hoods and cable entry seals inspected

Packing and shipping

- [ ] Desiccant inside, and openings sealed for the trip
- [ ] A temporary supply point for space heaters during storage
- [ ] Impact and tilt recorders fitted to transformers and large sections
- [ ] Lifting points, centre of gravity and weights marked on each split
- [ ] Delivery booked inside the winter road or barge window with margin

SHEET C · ON SITE, BEFORE ENERGIZATION

Receiving, Storage & Installation

Cold sites punish open enclosures and cold cables. Keep the equipment dry and warm from the day it arrives until it is energized.

PROJECT _____ EQUIPMENT / TAG _____ DATE _____ WITNESS _____

Receiving and storage

- [ ] Impact and tilt recorders read and photographed on arrival
- [ ] Equipment inspected for damage and moisture before acceptance
- [ ] Stored indoors, or covered, off the ground and with heaters energized
- [ ] Liquid-filled units checked for leaks and correct fluid level for the temperature

Installation

- [ ] Enclosures opened only inside a tent or hoarding with temporary heat
- [ ] Cable warmed above its minimum installation temperature before pulling
  Check the cable data sheet. Storing reels in a heated space for a day is a common approach.
- [ ] Conduit and cable entries sealed against cold air and condensation
- [ ] Anchors and foundations suited to frost heave or permafrost
- [ ] Bolted joints torqued and recorded; re-check plan agreed for spring

SHEET D · BEFORE AND DURING FIRST ENERGIZATION

Energization in the Cold

Run this after the normal pre-energization checklist. Moisture and cold fluid are the two things that make a cold start go wrong.

PROJECT _____ EQUIPMENT / TAG _____ DATE _____ WITNESS _____

Before closing in

- [ ] Space heaters running long enough to dry the equipment
  Agree the time with the manufacturer; a day or more is common after cold storage.
- [ ] Insulation resistance measured and compared with factory values; dry out if low
- [ ] Liquid-filled: fluid level and temperature recorded; cold-start guidance followed
- [ ] Breakers operated and timed at site temperature
- [ ] DC battery voltage and charger output checked in the cold
- [ ] Relay and HMI displays readable and responsive at site temperature

After energization

- [ ] Load built up gradually where the fluid or windings are very cold
- [ ] Heaters and thermostats confirmed working on permanent power
- [ ] Infrared scan once the equipment is carrying load
- [ ] All readings recorded for the winter baseline

SHEET E · EVERY YEAR, BEFORE FREEZE-UP AND THROUGH THE WINTER

Winter Operation & Maintenance

A short routine each autumn and a few checks through the winter keep cold-climate equipment reliable for decades.

PROJECT _____ EQUIPMENT / TAG _____ DATE _____ WITNESS _____

Before freeze-up

- [ ] Every space heater, thermostat and humidistat tested
- [ ] Door seals, vent hoods and filters inspected and replaced as needed
- [ ] Standby generator block heaters, batteries and fuel checked
- [ ] Spare heaters, fuses and relays on site and stored warm
- [ ] Winter access plan confirmed for emergency response

Through the winter

- [ ] Snow and ice cleared from vents, doors and pad-mounted cabinets
- [ ] Inside of enclosures checked for frost, condensation or tracking
- [ ] Liquid-filled fluid levels and temperatures logged
- [ ] Heater-failure alarms tested